

Todd Takala

Lead Data Scientist — Lead Machine Learning Engineer — AI/ML Engineer — Remote

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Summary of Qualifications

Lead Data Scientist with a proven track record of delivering over \$1.3 billion in business value through machine learning and end-to-end data science solutions to solve complex challenges. Manage complete project lifecycle, including stakeholder engagement, KPI definition, development, and deployment.

Experience

Lead Data Scientist — Caterpillar Inc. — Surprise, AZ

Sept 2022 – Present

- Led the end-to-end development of high-impact data science initiatives, including an award-winning ML system to predict equipment repair times. Engineered and deployed robust MLOps pipelines using AWS and Snowflake and established best practices for CI/CD and system architecture. Managed project lifecycles and created comprehensive monitoring plans to validate business impact.
- Engineered and deployed robust ETL/MLOps pipelines using Snowflake and AWS services (SageMaker, S3, Step Functions) to automate model retraining. Championed MLOps best practices, setting new standards for GitHub CI/CD, Docker, and AWS architecture.
- Spearheaded the creation of an ML/AI system to predict equipment repair times, resulting in an \$80 million reduction in warranty costs and a CEO Award.

Reliability Analytics Manager — Amazon — Tempe, AZ

Feb 2022 – Sept 2022

- Pioneered a new design-for-reliability benchmark by analyzing petabyte-scale IoT data from a large robotics fleet. Authored an influential white paper on complex Weibull analysis and secured adoption of the new benchmark by senior leadership.
- Analyzed over 2 petabytes of IoT data from a 400,000-robot fleet to establish a new design-for-reliability benchmark.
- Designed and optimized winning regression and recommendation systems, achieving two-time champion status in the Amazon Machine Learning University competition.

Data Engineering Manager / Reliability Manager — Empire-Cat — Mesa, AZ

Jan 2012 – Feb 2022

- Directed a high-performance team of up to 8 engineers, fostering a data-driven approach to solve equipment performance issues.
- Resolved critical design flaws in mining equipment through reproducible analysis in R and SQL, saving clients \$1.2 billion over six years. Created a regression model that identified key performance factors, leading to \$62 million in savings within 90 days.
- Led a team to establish an MLOps pipeline with ARIMA and SVM for AI-driven forecasting in supply chain management, integrated data, managed a component exchange program, collaborated with clients on durability forecasting, and addressed design flaws in mining equipment, reducing client costs by \$1.2B. This involved creating a comprehensive machine learning dataset from NLP on work orders and IoT telemetry.

Skills

AI/ML, Agile, Anomaly Detection, Architectural Design, Artificial Intelligence, AWS (ECR, Event Bridge, Redshift, Sagemaker, Step Functions), Business Requirements, Coaching, Continuous Integration, Creative, Database Administrator, Data Gathering, DBA, Deep Learning, DevOps, Developer Tools, Distributed Computing, Docker, Embedding, ETL, Git, GitHub Actions, GitHub Copilot, Growth Mindset, LLMs, Large Language Models, Mentoring, Problem Solving, Python (Autogluon, keras, numpy, pandas, PyTorch, Scikit-learn, TensorFlow), R (dplyr, prophet, tidyverse), Research, Snowflake, SQL, Streamlit, Statistical Tools, Strategic Thinking, Adaptability, Transformers, Willing to Learn

Education

Bachelor of Science in Mechanical Engineering — Arizona State University — Tempe, AZ 2006

STEM coursework included:

- Computational Thermodynamics, Design of Experiments (DoE), Differential Equations, Economics, Finance, Finite Element Analysis, Linear Algebra, Logic, Manufacturing, Material Science, Multivariate Calculus, Numerical Methods, Statistics

Certifications

Arizona State Board of Technical Registration

EIT, Mechanical Engineering – Registration No. 10670

BerkeleyX

The Science of Happiness at Work Professional Certificate

Canonical

Ubuntu Linux Professional Certificate

CSCMP

Supply Chain Foundations Demand Planning Professional Certificate

Data Camp

AI Fundamentals Certificate, Certified SQL Associate, Data Scientist Professional Certificate

Empire-Cat

Business Tools for Service Managers, Coaching for Peak Performance, Communicating for Leadership Success, Delegating with Purpose, Lean 6 Sigma Green Belt, Resolving Workplace Conflict, Setting Performance Expectations

GitHub

Career Essentials in GitHub Professional Certificate

IUOE Local 150

Diesel Mechanic, Heavy Equipment Operator (Bulldozers, Class B CDL, Cranes, Forklifts, MSHA, Preventive Maintenance, Scrapers, Skid-steers, Telehandlers, Wheel Loaders) Shielded Metal Arc Welding (SMAW)

Komatsu North America

Certified Technical Communicator

Society of Automotive Engineers

Cost and Finance for Engineers

Udemy

Streamlit for Snowflake Masterclass